Replicating Jegadeesh and Titman (1993): Momentum Portfolio Returns and Risk-Adjusted Performance

Abstract

This project examines the profitability of momentum investment strategies by replicating selected empirical analyses in Jegadeesh and Titman (1993), *Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency*. The study investigates whether stocks that have performed well in the past continue to outperform stocks that have performed poorly, and whether this return difference can be explained by firm characteristics or systematic market risk.

Using CRSP stock return data, the project constructs portfolios based on past stock performance and evaluates their subsequent returns. The empirical analysis focuses on Tables I–III of the original paper, including momentum portfolio returns, the performance of winner-minus-loser strategies, and portfolio returns within firm-size and beta subsamples. Overlapping formation and holding periods are used to reproduce the portfolio construction approach. Newey–West standard errors are applied when testing mean portfolio returns to account for serial dependence associated with overlapping holding periods.

The current implementation uses data covering 2000–2025 rather than the original 1965–1989 sample. Consequently, the analysis should be interpreted as a replication of the empirical methodology combined with an extension to a more recent period. The final conclusions will be completed after all portfolio-return and risk-adjustment estimates have been verified.

1. Introduction

Momentum is a well-documented empirical pattern in which stocks with relatively high past returns tend to continue outperforming stocks with relatively low past returns over intermediate horizons. Jegadeesh and Titman (1993) investigate this pattern by forming portfolios according to past returns and examining the performance of strategies that buy past winners and sell past losers.

The finding is important for asset pricing and market efficiency. If past stock returns contain information about future returns, a momentum strategy may earn positive returns. However, these profits do not necessarily imply market inefficiency: they

could partly reflect differences in systematic risk, firm characteristics, or the properties of stock returns.

This project addresses three questions:

1. Do momentum strategies generate positive average returns?
2. Does the profitability of momentum strategies remain evident when stocks are divided into firm-size groups?
3. Does the profitability remain evident when stocks are divided into groups based on their estimated market betas, and how does risk adjustment affect the results?

To answer these questions, the project reconstructs the portfolio formation and holding procedures used in the original study. It then examines the extent to which the results are consistent with the original findings in a more recent sample period.

2. Data

2.1 Stock return data

The project uses CRSP stock return data. The daily data-processing file, crsp_daily_common_base_2000_2025.dta, contains 36,867,762 observations and covers 2000–2025. The analysis retains common stocks using the security-subtype classification and removes exact duplicate records. The stock identifier is PERMNO, and the daily date variable is date.

The principal daily return variable is DlyRet. The variable DlyPrevCap records lagged market capitalization and is used as a weight when constructing the project's daily market-return proxy.

For the portfolio analysis, monthly stock returns are used to rank stocks, construct portfolios, and calculate holding-period returns. The monthly stock-level data are processed separately from the daily data used for beta estimation.

2.2 Market returns and risk-free rates

A daily market-return proxy is constructed by weighting individual stock returns by their lagged market capitalizations:

$$\frac{\sum_i \text{DlyPrevCap}\, i, tRi, t}{\sum_i \text{DlyPrevCap}_{i,t}}$$

The resulting series is saved in JT_market_daily_proxy.dta, with the variables mkt_ret, mkt_lag, and mkt_lead. The lagged and leading returns are aligned according to trading-day order.

Data limitation: this series is a value-weighted proxy constructed from the available common-stock data. It is not the official CRSP Value-Weighted Index used in the original paper. This distinction matters when interpreting the beta estimates and market-model results.

Monthly Fama–French factors are stored in JT_FF_factors_monthly.dta. The variables include the market excess return (mktrf), the size factor (smb), the value factor (hml), and the risk-free rate (rf). These variables have been converted from percentage units into decimal returns.

The Fama–French factors can support supplementary robustness analysis. However, a three-factor regression should not be described as an exact replication of the original Table III Panel B, which uses a market-model specification based on the CRSP Value-Weighted Index.

2.3 Sample-period comparability

The original paper studies the period January 1965 to December 1989. The current implementation uses a more recent period, with the available portfolio-return series extending from December 2001 through December 2025.

The difference in sample periods means that the estimates need not match the published numerical results. Changes in market structure, stock characteristics, trading practices, and the composition of listed firms may affect the measured profitability of momentum strategies.

Accordingly, the project evaluates whether the original empirical patterns are present in the more recent sample rather than treating numerical equality with the original paper as the sole criterion for success.

3. Methodology

3.1 Portfolio formation and momentum strategy

Following Jegadeesh and Titman (1993), stocks are ranked according to their cumulative returns over a prior formation period. Stocks are then allocated to ten decile portfolios, from the lowest past-return decile (P1) to the highest past-return decile (P10).

The principal six-month formation and six-month holding strategy, denoted J=6 and K=6, uses past six-month returns to rank stocks and holds the resulting portfolios for six months. Portfolio returns are calculated using equal weights across the constituent stocks.

The winner-minus-loser strategy is defined as

$$R_{\mathrm{WML},t} = R_{P10,t} - R_{P1,t}$$

where ($R_{P10,t}$) is the return on the winner portfolio and ($R_{P1,t}$) is the return on the loser portfolio. A positive WML return indicates that past winners outperform past losers during the holding period.

To increase the number of usable portfolio observations, the strategy employs overlapping holding periods. Each month, a new portfolio is formed while portfolios formed in previous months remain active until their holding periods end. Monthly portfolio returns are obtained by averaging the returns of the active portfolios.

The implementation also considers alternative formation and holding horizons of 3, 6, 9, and 12 months. Where applicable, a second specification skips the first five trading days of a newly formed portfolio to reduce the influence of short-term price pressure and related effects.

3.2 Statistical inference

The main hypothesis concerns the average return of the winner-minus-loser strategy:

$$H_0(R_{\mathrm{WML}}) = 0$$

against the alternative

$$H_1(R_{\mathrm{WML}}) \neq 0$$

The mean return is estimated from the monthly WML return series. Statistical significance is evaluated using Newey–West standard errors, with the lag length set to K−1 for a K-month holding strategy. For the six-month holding strategy, this corresponds to five lags.

This adjustment is important because overlapping portfolios can induce serial dependence in monthly strategy returns. Conventional standard errors that ignore this dependence may produce misleading significance tests.

3.3 Table I: Momentum strategy returns

Table I evaluates the average returns of winner, loser, and winner-minus-loser portfolios across different formation and holding horizons.

For each specification, the analysis reports the mean monthly return, the associated Newey–West t-statistic, and the number of monthly observations. The comparison across horizons helps establish whether the momentum effect is concentrated at particular formation or holding periods.

The results will be interpreted by considering both the magnitude of the WML return and its statistical significance. The skipped-week specification is compared with the corresponding baseline specification to assess whether the results are sensitive to excluding the first five trading days of a new portfolio.

3.4 Table II: Characteristics of momentum portfolios

Table II examines the characteristics of the portfolios formed using past six-month returns. In addition to portfolio returns, the analysis considers the average market capitalization of stocks assigned to each decile.

This comparison helps establish whether winner and loser portfolios differ systematically in firm size. Such differences matter because size may be associated with expected returns, market risk, and trading frictions.

The average capitalization measures are constructed from the capitalization information available at portfolio formation. Their units must be interpreted according to the units of the underlying CRSP capitalization variable.

3.5 Table III: Size and beta subsamples

Table III investigates whether momentum profitability persists when the investment strategy is applied to stocks within more homogeneous groups.

Size portfolios. Stocks are divided into three size groups: S1 (small firms), S2 (medium-sized firms), and S3 (large firms). Within each size group, stocks are ranked according to their past six-month returns and allocated to ten deciles. The WML return is then calculated separately for each size group.

Beta portfolios. Stocks are also divided into three groups according to their estimated Scholes–Williams betas: low-beta, medium-beta, and high-beta stocks. The original paper forms these groups using beta estimates from the calendar year preceding portfolio formation.

The current implementation estimates daily stock-level regressions of the form

$$R_{i,t} = a_i + b_{i,-1}R_{m,t-1} + b_{i,0}R_{m,t} + b_{i,+1}R_{m,t+1} + u_{i,t}$$

The Scholes–Williams beta is calculated as

$$\frac{\hat{b}_{i,-1} + \hat{b}_{i,0} + \hat{b}_{i,+1}}{1 + 2\hat{\rho}_m}$$

where ($\hat{\rho}_m$) is the estimated first-order autocorrelation of the market-return proxy. This adjustment accounts for potential nonsynchronous trading in daily stock returns.

The current Stata implementation requires at least 100 valid daily observations in each stock-year regression. This is a practical data-quality threshold and should be disclosed as an implementation choice rather than attributed to the original paper unless independently verified.

Panel A reports average monthly portfolio returns and WML returns within the full sample and the size- and beta-based subsamples.

Panel B evaluates risk-adjusted performance using a market-model benchmark. The intercept from the portfolio-return regression represents the estimated abnormal return under the specified benchmark. Because the current market-return series is a constructed proxy rather than the official CRSP Value-Weighted Index, the resulting estimates are best described as market-proxy-adjusted returns unless the official index is obtained.

The principal question is whether positive WML returns remain evident within size and beta groups. If they do, the findings would suggest that momentum profitability is not confined to a single size category or beta category. The final conclusion will depend on the estimated returns and their statistical significance.

4. Empirical Results

This section will be completed using the final Stata output. Numerical estimates and significance claims are not inserted until the corresponding output has been checked.

4.1 Table I results

The final Table I results will be used to assess the average returns of P1, P10, and WML across the different formation and holding horizons. The analysis will report whether the WML returns are positive and statistically significant, and whether the results differ between the baseline and skipped-week specifications.

J	K	panel	leg	mean_monthly	t_stat	N
3	3	A	Buy	0.008954	2.028343	289
3	3	A	Sell	0.004834	0.76062	289
3	3	A	WML	0.00412	1.192992	289
3	6	A	Buy	0.009468	2.006376	289

3	6	A	Sell	0.004213	0.644376	289
3	6	A	WML	0.005255	1.647376	289
3	9	A	Buy	0.009787	2.081377	289
3	9	A	Sell	0.003912	0.570995	289
3	9	A	WML	0.005875	1.714815	289
3	12	A	Buy	0.009347	2.015718	289
3	12	A	Sell	0.004046	0.58445	289
3	12	A	WML	0.005301	1.598435	289
6	3	A	Buy	0.011727	2.740811	289
6	3	A	Sell	0.00453	0.659336	289
6	3	A	WML	0.007198	1.661008	289
6	6	A	Buy	0.011363	2.574379	289
6	6	A	Sell	0.003746	0.519665	289
6	6	A	WML	0.007618	1.673702	289
6	9	A	Buy	0.010761	2.475024	289
6	9	A	Sell	0.003561	0.474953	289
6	9	A	WML	0.0072	1.532456	289
6	12	A	Buy	0.00998	2.31774	289
6	12	A	Sell	0.004013	0.532006	289
6	12	A	WML	0.005967	1.292318	289
9	3	A	Buy	0.01264	3.119281	289
9	3	A	Sell	0.004453	0.60475	289
9	3	A	WML	0.008187	1.604223	289
9	6	A	Buy	0.011238	2.701751	289
9	6	A	Sell	0.003842	0.500726	289
9	6	A	WML	0.007396	1.402405	289
9	9	A	Buy	0.010453	2.545472	289
9	9	A	Sell	0.003988	0.499977	289
9	9	A	WML	0.006465	1.17203	289
9	12	A	Buy	0.009821	2.395091	289
9	12	A	Sell	0.004335	0.544047	289
9	12	A	WML	0.005486	1.005961	289
12	3	A	Buy	0.011085	2.770705	289
12	3	A	Sell	0.004658	0.626295	289
12	3	A	WML	0.006427	1.232476	289
12	6	A	Buy	0.01015	2.526024	289
12	6	A	Sell	0.004564	0.580169	289
12	6	A	WML	0.005586	0.992983	289
12	9	A	Buy	0.009696	2.420961	289
12	9	A	Sell	0.00447	0.547454	289
12	9	A	WML	0.005226	0.885315	289
12	12	A	Buy	0.009281	2.315969	289
12	12	A	Sell	0.005049	0.620486	289
12	12	A	WML	0.004232	0.726939	289

3	3	B	Buy	0.008224	2.096027	289
3	3	B	Sell	0.003022	0.532279	289
3	3	B	WML	0.005202	1.67631	289
3	6	B	Buy	0.009103	2.060221	289
3	6	B	Sell	0.003307	0.533481	289
3	6	B	WML	0.005796	1.925275	289
3	9	B	Buy	0.009544	2.12658	289
3	9	B	Sell	0.003307	0.499698	289
3	9	B	WML	0.006236	1.886524	289
3	12	B	Buy	0.009164	2.046648	289
3	12	B	Sell	0.003593	0.532009	289
3	12	B	WML	0.005572	1.722184	289
6	3	B	Buy	0.011011	2.926173	289
6	3	B	Sell	0.002494	0.40465	289
6	3	B	WML	0.008517	2.162846	289
6	6	B	Buy	0.011005	2.681378	289
6	6	B	Sell	0.002728	0.398521	289
6	6	B	WML	0.008278	1.906846	289
6	9	B	Buy	0.010522	2.548173	289
6	9	B	Sell	0.002882	0.397784	289
6	9	B	WML	0.00764	1.675453	289
6	12	B	Buy	0.009801	2.364238	289
6	12	B	Sell	0.003504	0.476271	289
6	12	B	WML	0.006297	1.39267	289
9	3	B	Buy	0.011735	3.329016	289
9	3	B	Sell	0.002303	0.348761	289
9	3	B	WML	0.009432	2.047727	289
9	6	B	Buy	0.010785	2.805407	289
9	6	B	Sell	0.002767	0.3804	289
9	6	B	WML	0.008018	1.598014	289
9	9	B	Buy	0.010151	2.612462	289
9	9	B	Sell	0.003271	0.424876	289
9	9	B	WML	0.00688	1.286614	289
9	12	B	Buy	0.009594	2.435978	289
9	12	B	Sell	0.003797	0.489131	289
9	12	B	WML	0.005797	1.086795	289
12	3	B	Buy	0.010289	2.946803	289
12	3	B	Sell	0.002573	0.386548	289
12	3	B	WML	0.007716	1.643833	289
12	6	B	Buy	0.009752	2.621031	289
12	6	B	Sell	0.003521	0.473197	289
12	6	B	WML	0.00623	1.16828	289
12	9	B	Buy	0.00943	2.482756	289
12	9	B	Sell	0.003775	0.480101	289

12	9	B	WML	0.005655	0.992803	289
12	12	B	Buy	0.009082	2.355629	289
12	12	B	Sell	0.004527	0.572281	289
12	12	B	WML	0.004554	0.803328	289

Table I reports the average monthly returns of the winner (Buy), loser (Sell), and winner-minus-loser (WML) portfolios across different formation and holding periods in Panels A and B. The average monthly returns of the Buy portfolios are consistently positive and generally higher than those of the Sell portfolios. In contrast, the Sell portfolios exhibit relatively low average returns, with t-statistics below one in all reported cases.

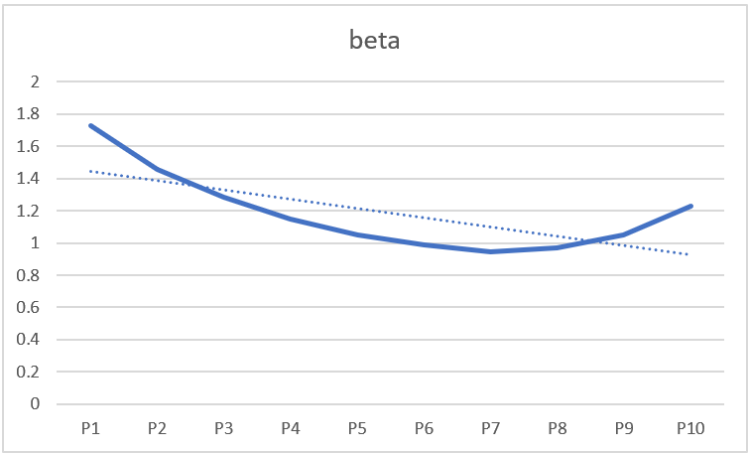
The WML portfolios generate positive average monthly returns across all specifications, ranging from approximately 0.41% to 0.94%. This pattern is consistent with the presence of momentum effects, as stocks with stronger past performance tend to outperform those with weaker past performance. However, most WML portfolios do not achieve conventional levels of statistical significance. The strongest result appears in Panel B for the (J=9, K=3) strategy, which earns an average monthly return of approximately 0.94% with a t-statistic of 2.05.

Overall, the results provide suggestive, but not uniformly strong, evidence of momentum profits. The positive return differentials are economically meaningful in magnitude, although their statistical significance varies across portfolio specifications and panels.

4.2 Table II results

portfolio	decile	beta	avg_cap_millions	alpha	r2	N	avg_cap_raw
P1	1	1.727084	592.2719	-0.01127	0.53666	289	592271.9
P2	2	1.456996	2016.327	-0.00548	0.715657	289	2016327
P3	3	1.282072	3788.077	-0.00283	0.793733	289	3788077
P4	4	1.150731	5531.754	-0.00123	0.831068	289	5531754
P5	5	1.052785	6849.831	0.000717	0.84979	289	6849831
P6	6	0.990162	7549.812	0.001195	0.858354	289	7549812
P7	7	0.94665	7955.185	0.002216	0.834613	289	7955185
P8	8	0.971013	8147.642	0.002138	0.827412	289	8147642
P9	9	1.052913	7571.126	0.002145	0.811075	289	7571126
P10	10	1.229838	3769.584	0.000669	0.705636	289	3769584
P10-P1		-0.49725		0.011942	0.108738	289	

The final Table II results will be used to describe how portfolio characteristics vary across the ten past-return deciles. Particular attention will be paid to the capitalization of the winner and loser portfolios and to whether the observed pattern suggests systematic differences in firm size.



The market betas exhibit a broadly U-shaped pattern across the ten portfolios, declining from P1 to P7 before increasing toward P10. The average market capitalization generally increases from P1 to P8 but decreases for P9 and P10. The estimated alphas are negative for P1–P4 and positive for P5–P10. The P10–P1 portfolio has a negative beta of −0.497 and a positive alpha of 0.011942, suggesting a positive return differential after accounting for market exposure under the specified regression model.

4.3 Table III results

The size-subsample results will be compared across S1, S2, and S3. The beta-subsample results will be compared across the low-, medium-, and high-beta groups. Both the raw WML returns and the market-model-adjusted estimates will be considered.

The analysis will distinguish between economic magnitude and statistical significance. A positive estimated return does not, by itself, establish that the return is statistically different from zero. Likewise, failure to reject a zero-return hypothesis does not prove that the true return is zero.

sample	sizegrp	portfolio	alpha	nw_t	N	portfolio_num	is_WML
All	0	P1	-0.01035	-2.22034	289	1	0
All	0	P2	-0.00489	-1.89741	289	2	0
All	0	P3	-0.00247	-1.40045	289	3	0
All	0	P4	-0.00105	-0.6842	289	4	0
All	0	P5	0.000769	0.5508	289	5	0
All	0	P6	0.001162	0.86123	289	6	0

All	0	P7	0.002123	1.440179	289	7	0
All	0	P8	0.002083	1.41456	289	8	0
All	0	P9	0.002201	1.353321	289	9	0
All	0	P10	0.000966	0.438939	289	10	0
All	0	WML	0.011317	2.940303	289	11	1
Small	1	P1	-0.01316	-2.03136	289	1	0
Small	1	P2	-0.00616	-1.34003	289	2	0
Small	1	P3	-0.00504	-1.36362	289	3	0
Small	1	P4	-0.00263	-0.83994	289	4	0
Small	1	P5	-0.00077	-0.28153	289	5	0
Small	1	P6	0.001868	0.736594	289	6	0
Small	1	P7	0.003758	1.543224	289	7	0
Small	1	P8	0.004377	1.634129	289	8	0
Small	1	P9	0.006214	2.099939	289	9	0
Small	1	P10	0.001507	0.458191	289	10	0
Small	1	WML	0.014671	3.108537	289	11	1
Medium	2	P1	-0.01026	-2.75476	289	1	0
Medium	2	P2	-0.00446	-2.04912	289	2	0
Medium	2	P3	-0.00272	-1.5917	289	3	0
Medium	2	P4	-0.00117	-0.7041	289	4	0
Medium	2	P5	0.000172	0.105922	289	5	0
Medium	2	P6	0.000865	0.516862	289	6	0
Medium	2	P7	0.001878	1.041114	289	7	0
Medium	2	P8	0.001412	0.774774	289	8	0

Medium	2	P9	0.001121	0.626142	289	9	0
Medium	2	P10	0.000298	0.126013	289	10	0
Medium	2	WML	0.010555	3.043107	289	11	1
Large	3	P1	-0.00654	-2.95963	289	1	0
Large	3	P2	-0.00263	-2.25369	289	2	0
Large	3	P3	-0.00043	-0.42032	289	3	0
Large	3	P4	0.000176	0.173867	289	4	0
Large	3	P5	0.000692	0.746417	289	5	0
Large	3	P6	0.000689	0.732469	289	6	0
Large	3	P7	0.000937	1.107689	289	7	0
Large	3	P8	0.000717	0.779139	289	8	0
Large	3	P9	0.000469	0.479434	289	9	0
Large	3	P10	0.000916	0.594602	289	10	0
Large	3	WML	0.00746	2.650915	289	11	1

Table III, Panel B reports the market-model alphas of the ten momentum portfolios (P1–P10) and the winner-minus-loser (WML) portfolio for the full sample and three size groups. The regressions use monthly excess portfolio returns as the dependent variable and market excess returns as the explanatory variable. Newey–West standard errors with five lags are used to account for potential heteroskedasticity and autocorrelation.

For the full sample, the alpha of the loser portfolio (P1) is negative and statistically significant at the 5% level (−0.01035, t = −2.22), whereas the alpha of the winner portfolio (P10) is positive but statistically insignificant (0.00097, t = 0.44). In contrast, the WML portfolio has a positive alpha of 0.01132 per month, which is statistically significant at the 1% level (t = 2.94). This finding suggests that the winner-minus-loser strategy earns a positive risk-adjusted return that cannot be explained by its exposure to the market factor alone. The result is driven primarily by the significantly negative alpha of the loser portfolio rather than by a statistically significant positive alpha of the winner portfolio.

The size-sorted results show a similar pattern. The WML alphas are positive and statistically significant at the 1% level for small-, medium-, and large-size portfolios, with monthly estimates of 0.01467 (t = 3.11), 0.01056 (t = 3.04), and 0.00746 (t = 2.65), respectively. The estimated momentum return is largest among small firms and decreases across the medium- and large-size groups. This pattern is consistent with the possibility that the momentum effect is stronger among smaller firms. However, the differences in WML alphas across size groups cannot be concluded to be statistically significant without a formal test of equality.

Overall, the results provide evidence of a positive momentum effect in the sample, even after controlling for market risk. Nevertheless, the findings should be interpreted as evidence for the sample period studied rather than as a direct replication of the original estimates in Jegadeesh and Titman (1993).

5. Discussion and Limitations

Several limitations should be considered when interpreting the estimates.

First, the sample period differs from that of the original paper. Results from 2000–2025 cannot establish that the published numerical estimates have been exactly reproduced for 1965–1989. Instead, they show how the strategy performs in the more recent period.

Second, the market-return proxy is constructed from the available CRSP common-stock sample. It may differ from the official CRSP Value-Weighted Index in its coverage and weighting. This affects the comparability of beta estimates and market-model intercepts with those in the original paper.

Third, the annual Scholes–Williams beta estimates depend on the availability of valid daily returns. The minimum-observation requirement improves consistency across stock-year regressions but may exclude less frequently observed securities. The resulting beta portfolios may therefore differ from those in the original study.

Fourth, portfolio construction and inference depend on the correct alignment of formation months, holding months, and overlapping vintages. The final reproducibility checks should verify that each portfolio uses only information available at formation, that monthly returns are aligned correctly, and that the reported sample size matches the estimation sample.

Finally, Fama–French factor regressions can provide supplementary evidence on risk-adjusted performance, but they are not a substitute for reproducing the original market-model specification. Any supplementary factor analysis should be reported separately.

6. Conclusion

This project applies the portfolio formation and return-testing framework of Jegadeesh and Titman (1993) to a more recent CRSP sample. It constructs winner and loser portfolios based on past stock returns, evaluates the winner-minus-loser strategy, and examines whether the results persist within firm-size and beta subsamples.

The final assessment will be based on the completed Table I–III estimates. In particular, it will consider whether WML returns are positive and statistically significant, whether the results vary across size and beta groups, and whether market-model adjustment materially changes the conclusions.

The study contributes a reproducible implementation of the momentum strategy and an extension of the original empirical design to a later period. Its conclusions will be qualified by the differences in sample period and market benchmark.

References

Jegadeesh, N., & Titman, S. (1993). Returns to buying winners and selling losers: Implications for stock market efficiency. *The Journal of Finance, 48*(1), 65–91.

Newey, W. K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica, 55*(3), 703–708.

Scholes, M., & Williams, J. (1977). Estimating betas from nonsynchronous data. *Journal of Financial Economics, 5*(3), 309–327.

Reproducibility and AI-use disclosure

The empirical analysis is implemented in Stata. The reproducibility package should include the do-files used to clean the data, construct portfolios, estimate beta coefficients, perform statistical tests, and export the final tables. It should also document the input files, sample restrictions, variable definitions, and output filenames.

AI assistance was used to support code development, debugging, methodological explanations, and the drafting and organization of this report. The data processing, final code execution, numerical results, interpretation of statistical output, and verification of the submitted materials are checked by the author.